CyBOK

Digital forensics

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Agenda

- Cyber in crime
- Digital forensics
- Forensic examination
- Reporting
- Autopsy demo

Litterature note

- This lecture is based on the following CyBOK knowledge areas:
  - Forensics

Cybercrime

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- A very common misconception is that computer forensic experts are working on cybercrimes.
- That misconception makes it useful to discuss cybercrime and when digital evidence is useful in criminal investigations.
- First, what constitutes a cybercrime is not really interesting on the topic of computer forensics, computer forensics deal with digital evidence and that is present in any type of crime.

Cybercrime

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- However, when discussing cyber crime one could argue that it is about crimes against computers, such as hacking attacks.

- Rogers (2001) involved several types of criminals in his discussion on computer criminals, including hackers and fraudsters

- Discussing crimes in terms of means motives and opportunity, as Rogers (2000), one could argue that a cyber crime is a crime where the means is computerized.

Cybercrime

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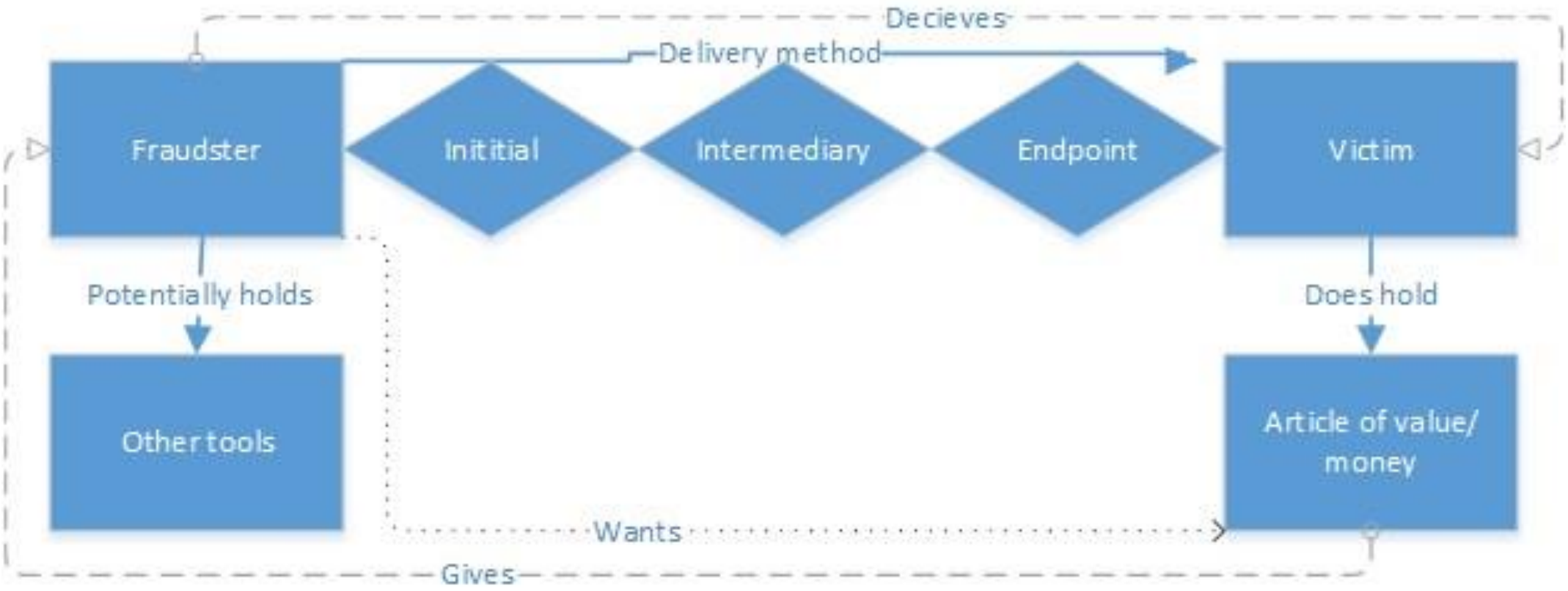
- Perhaps a more useful discussion would be to discuss crimes in terms of pure cybercrimes and computer aided crimes.
  - Cybercrimes would involve crimes that are completely cyber, such as intrusions, denial of service and so on
  - Computer aided crimes would involve traditional crimes that are aided by computers, such as online frauds or online drug trade
  - Offline crimes would be acts that does not rely on or are aided by computers in any way
- However, digital evidence can still play a role in any type of crime!

Example - (online) fraud



Cyber enabled and dependent crime

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- Modern terminology talks about cyber-enabled and dependent crime
- Cyber-enabled crime is a traditional crime that utilizes cyber
- Cyber-dependent crime is a crime that depends on cyber for its execution

REFLECT!

- Use one minute to write down one-liners which describe what you just learned – come up with as many as you can.
- Join up in pairs or groups and compare your lists for another minute.
- Questions so far?

Digital forensics

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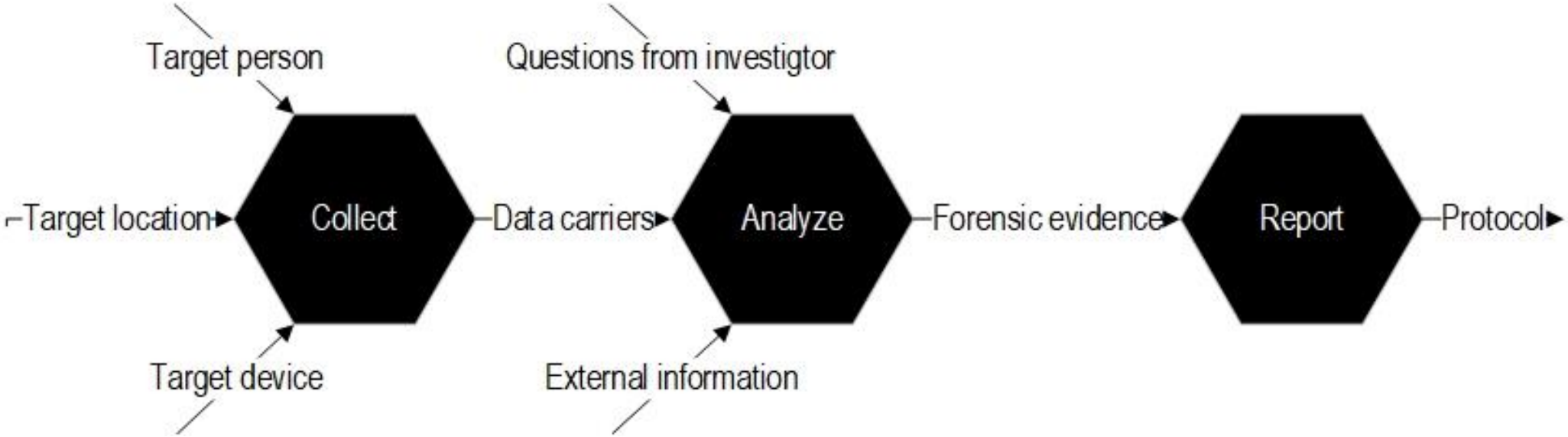
- The foundation of digital forensics is that it is the practice of collecting, analyzing and reporting on digital data.
  - Combined, that is the process of a forensic examination

- Implicit is that there is someone who is requesting the examination, and that someone must have a reason for wanting the examination.

- There is also someone who is examined, or who's devices are examined

Digital forensics



Digital forensics – DFRWS definition

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Digital forensics is the use of scientifically derived and proven methods toward the preservation, collection, validation, identification, analysis, interpretation, documentation and presentation of digital evidence derived from digital sources for the purpose of facilitating or furthering the reconstruction of events found to be criminal, or helping to anticipate unauthorised actions shown to be disruptive to planned operations.

Digital forensics – NIST definition

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Digital forensics (NIST) is considered the application of science to the identification, collection, examination, and analysis of data while preserving the integrity of the information and maintaining a strict chain of custody for the data. Data refers to distinct pieces of digital information that have been formatted in a specific way.

Digital forensics – CyBOK definition

Digital forensics is the process of identifying and reconstructing the relevant sequence of events that have led to the currently observable state of a target IT system or (digital) artifacts.

Digital forensics

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- Forensics is about reconstructing what happened in a digital environment
- Forensics is also extended to analyze what is currently happening
- Computer forensics is used in law enforcement as well as in corporate environments
  - Crime investigations
  - Civil cases
  - Intrusion detection/response
  - misc

Digital forensics

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- Digital forensics is about uncovering what has been done using a digital device or happened to it.
  - includes uncovering what is stored on the actual device
  - track internet history such as browsing and chat history
  - Analyze who authored a document or where a picture was taken
  - Identify information about intrusions.

- Basically, everything done with a digital device will leave prints that a skilled forensic expert can uncover.

Digital evidence

- Digital forensics is about examining digital evidence
  - Data of evidentiary value is commonly refereed to as artifacts

- Digital evidence is data collected from digital storage, that is subject for a forensic examination.

- From a technical viewpoint, the evidence collection process can be divided into handling devices that are on and devices that are off.
  - Dead box analysis vs. Live forensics

Digital evidence

- Whenever handling evidence, especially in law enforcement, it is of extreme importance that the state of the evidence is not changed
  - The evidence must be accepted by courts

- In computer forensics this means that the content of the storage media must not be changed at all.

- Since all computer usage will almost certainly leave traces and can thus modify the data on the storage device, actions must be taken to prepare a device for secure examination.

Digital evidence

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- Before an analysis can start, digital evidence must fall into the hand of the examiner
- Most common case is that a device arrives, seized and turned off
  - Dead box analysis
- In some cases the examiner gets the chance to start the examination from a running machine
  - Live forensics

Types of digital evidence

- Files and folders
  - What You See Is What You Get
- Deleted files which can be recovered
- Settings and logs
  - Log files in various formats
  - Windows registry
- Metadata
  - Data describing other data
  - Time stamps, permissions, etc

Dead box analysis

- Connect the storage device of the target computer to your own computer using a write blocker.
  - A write blocker prohibits your computer from writing data to the device and can be a software or a hardware device

- Create a forensic disk image of the device
  - A physical disk image is a bit-by-bit copy of the device
  - A logical disk image is a copy of one or more full partitions as seen by the operating system

- Work your case by analyzing the forensic disk image

- If it is impossible to create an image, you will need to work on the actual drive. You may even be forced to conduct the examination on the actual computer that is subject for examination. In these cases it is important that you understand that your actions can contaminate the evidence. Take careful notes!

Live forensics

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- Live forensics is about examining devices that are on.

- Live examinations introduces considerations regarding the integrity of the evidence. However, it allows you to collect volatile data:
  - Document what is currently on screen
  - Information about the current user, current network connections etc.
  - Memory dump
  - Analyze if any encryption software is used, do malware analysis etc

- The power of the memory analysis is that evil actions must be in their true form as far as low level instructions go.

- After the Live examination, the computer is often brought to the lab for dead box analysis

REFLECT!

- Use one minute to write down one-liners which describe what you just learned – come up with as many as you can.
- Join up in pairs or groups and compare your lists for another minute.
- Questions so far?

Analysis

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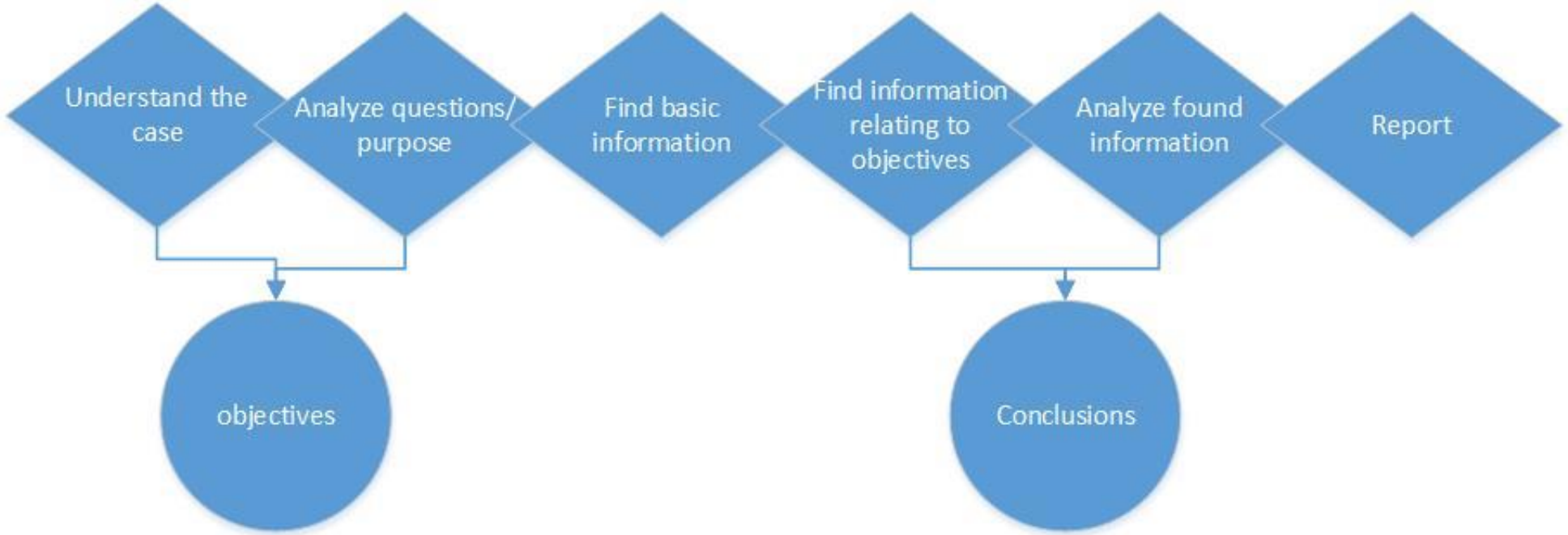
- A forensic examination is about answering questions asked by someone else. The common idea is to uncover what the computer has been used for, or what happened to it in order to prove or disprove something.
  - Malware / incident detection
  - Attribution

- A forensic examination must be
  - Unbiased
  - Reproducible
  - Comply with local laws and regulations.

- In a corporate environment, the requirements may be more loose

Analysis


Understand the case
Analyze questions/ purpose
Find basic information
Find information relating to objectives
Analyze found information
Report
objectives
Conclusions

Analysis

- Before you get going, you should spend some time understanding the case and the objectives handed to you to understand what you are asked to look for.

- This can involve:
  - Reading case description
  - Reading interrogation reports
  - Reading IDS logs
  - Questioning users

- The result should be objectives describing what to look for during the examination

Analysis – basic information examples

- *Account for all data* meaning that you examine the physical size of the hard drive and then look at how much of this space that is allocated to a visible partition.
- *Get computer install date, operating system version, list of users and registered owner* so that you easily answer questions about what operating system the computer used, how many users there was on the system and so on.
- *Get time zone information and clock settings* for the sole reason that time is commonly an important factor.
- *Find network drive maps* so that you can know if the computer has been connected to remote storage media.

Analysis

- Look for artifacts concerning the objectives of your examination
- Analyze the artifacts that you found and use this information to draw conclusions.
  - You may include external information when you draw conclusions
- Summarize your findings and conclusions in a report, this is discussed next.
- In incident response, it is common to go directly to the analysis for time purposes

REFLECT!

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Reporting

- Your findings should be presented in a report.
- Reflect a fair and objective view of your examination
- The local laws and regulations will dictate the content and format of the report.
- It is common that the report will be read by persons who are not computer experts – use appropriate language.

Reporting – typical content

- Case data
  - Data such as case identifier and evidence identifiers
- Purpose of examination
  - Description of the objectives of the examination
- Findings
  - The artifacts found during the examination
- Conclusions
  - Your conclusions based on the examination. Clearly separated from the findings and written with a common scale.

Autopsy

- Open source computer forensics tool

- Allows you to view and process digital evidence
  - View as in browsing in different ways
  - Process as in automatically perform various analysis tasks

- Lets have a look!

Thank you for today!